Magic Nutrition Hub

Complete Feature Inventory & Capability Overview

An end-to-end multi-tenant platform that runs every operational surface of a nutrition club — from member sign-up and face-recognition check-in, to live kitchen displays, wallet & billing, MLM/referral networks, body-transformation challenges, and a centralised super-admin command centre.

Portals	Public Landing • Reception • Kitchen Display • TV Display • Member • Staff • Owner • Super Admin
Mobile Apps	4 dedicated Android APKs — Reception, Kitchen, Admin, Landing (built with Capacitor)
Stack	React + TypeScript • Firebase (Auth, Firestore, Functions, FCM, Storage) • face-api.js • RawBT thermal printing
Highlights	Real-time sync • Face-recognition check-in • PWA + offline • Push notifications • MLM tree • Receipt printing

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1. Public Surfaces

The first touch points for visitors and prospective members.

1.1 Landing Page

Route: / · Public, unauthenticated

- Custom branded landing page per club (configurable HTML, club logo & colours).
- Quick-link hub to Reception, Member Login, Admin, Kitchen, TV.
- Auto-redirects logged-in users to their role-specific dashboard.
- Available as a dedicated landing.apk for kiosk deployment.

1.2 Join Page

Route: /join · Public self-service signup

- Capture: name, phone, WhatsApp, email, password, DOB, current/target weight, health conditions.
- Referral lookup — searchable by member name / ID / phone (auto-attaches to the MLM tree).
- Auto-generates a unique Member ID using a name-prefix scheme.
- Creates Firebase Auth user + Firestore member record + initial wallet in one step.

1.3 Leave-a-Review Page

Route: /leave-review · Public

- Star rating + free-text feedback form.
- Reviews appear instantly in the Owner’s Reviews dashboard for moderation.

2. Reception Kiosk

Route: /reception · 6-digit PIN gate · Reception APK

Tablet-mounted at the front desk. The complete check-in & ordering flow happens here.

- **Member check-in via face scan** — face-api.js (TinyFaceDetector + Landmark68 + Recognition) with liveness check.
- **123 / ABC keyboard toggle** for fast member & volunteer search.
- **Volunteer check-in** by QR code or by tapping a photo card on the volunteer grid.
- **Real-time volunteer photo grid** showing exactly who is on-shift right now.
- **Visitor log** with photo capture at every check-in.
- **Shake / Afresh order entry** on the spot — flows in real time to the Kitchen tab.
- **Receipt printing** via RawBT thermal printer (branded header, tight label rows).
- Performance: parallel writes + navigate-first + club print cache for instant check-in.
- PIN fallback if the camera or scanner fails.

3. Kitchen Display System (KDS)

Route: /kitchen · 6-digit PIN gate · Kitchen APK

Wall-mounted screen for the kitchen team. Driven by live Firestore listeners.

- **Live order queue** with member photos updating in real time.
- **Order workflow:** Pending → Ready → Served / Rejected.
- **Today's Specials** management — toggle, set limited / unlimited stock and quantity.
- **Kitchen stock dashboard** with In-Stock / Low / Critical badges per recipe item.
- **Stock request form with face verification** for accountant / manager approval.
- **Daily shake & Afresh order summary** with auto-print.
- **Recipe / menu editor** with full ingredient breakdown.
- **Auto-print countdown** — 10s timer before automatic receipt dispatch.
- Push notification fires to the member when their order is marked Ready.

4. TV Display

Route: /tv · Fullscreen carousel for in-club Smart TV

A rotating presentation that turns the screen on the wall into a live engagement channel.

- **Today’s Specials** — image + price slides.
- **Live order queue** — visual ranking of who’s next.
- **Recently served members** celebration slides.
- **Announcements** — text + images broadcast from the Owner portal.
- **Birthday & anniversary celebrations** with member photos.
- **Early Bird leaderboard** photo posts.
- **Fun Friday** engagement screen.
- **Club stats snapshot** + rotating inspirational quotes.
- Slide timing: 10s default, 30s on the queue slide. Welcome spin animation on load.

5. Owner Portal

Route: /owner/ · Firebase Auth (clubOwner role) + 8-digit PIN*

The club owner’s daily command centre — 17 dedicated modules covering every operational surface.

5.1 Dashboard

- At-a-glance: today’s attendance, pending orders, low stock alerts, expiring members.
- Pending top-up requests inline (approve / reject without leaving the page).
- Drastic weight changes alert (members with ±5% weekly change).
- “Today’s report” one-click print button.

5.2 Members

- Searchable members list with photos, plan, expiry & status.
- Member profile: weight progress, attendance history, MLM upline / downline, transactions.
- Edit any field, reset password, toggle active status.

5.3 Attendance

- Daily log with photos and timestamps.
- Volunteer grid for quick tap-to-check-in.
- Streaks & monthly summary analytics.

5.4 Orders Management

- Filter by date / member / status; bulk mark-ready and mark-served.
- Receipt printing per order; revenue and top-items stats.

5.5 Inventory

- Master product list, unit-based tracking (not packs).
- Reorder thresholds with traffic-light status colours.
- Face-verified restock workflow — basket then batch verify.

5.6 Expiry Management

- List of members expiring this week.
- Auto-renewal toggle per plan.
- Push reminders 7 days before expiry; printable expiry labels.

5.7 Billing & Reports

- Per-club Firebase usage cost estimate (INR).
- Invoice history + payment status badges.
- Custom report builder: attendance, revenue, inventory, growth — exportable to CSV.

5.8 Membership Plans

- Create / edit plans (credits, price, validity, auto-renew).
- Activate or deactivate plans without deleting history.

5.9 Wallet Approvals

- Pending top-up queue with full member context.
- Approve / reject with admin notes; bulk top-up to many members at once.

5.10 Volunteers

- Add volunteers, assign roles (accountant / inventory / kitchen / reception).
- Hour tracking via clock-in/out; weekly summary export.

5.11 Downline Clubs

- Sub-club hierarchy view for franchise / multilevel networks.
- Performance roll-up & commission tracking.

5.12 Menu

- Shake recipe editor: photos, ingredients, cost, sell price.
- One-tap availability toggle.

5.13 Leaderboard

- Members ranked by visits, weight loss, BTC progress, streak, network size.
- Exportable for printing or display.

5.14 BTC (Body Transformation Challenge)

- Multi-session tracking (baseline + weekly).
- Captures weight, body-fat %, BMR, muscle mass.
- Rankings + CSV bulk import.

5.15 Reviews

- Public review feed with moderation tools.

5.16 Announcements

- Text + image announcements with push-notification broadcast.
- Schedule for future publish; view delivery / engagement.

5.17 Club Settings

- Branding (logo, colour, currency name).
- Custom landing-page URL.
- PIN rotation for kitchen, reception, owner.
- Per-club shake / refreshment token toggles.
- Backup & restore of club data.

6. Staff Portal

Route: /staff/ · Firebase Auth (staff role)*

6.1 Staff Dashboard

- Clock in / out with elapsed shift time.
- Today’s attendance log.
- Manual member check-in (scanner backup).
- Quick member lookup by name or phone.

6.2 Order Entry

- Member lookup → select shake → submit to kitchen.
- Pending orders list visible to staff.

7. Member Portal

Route: /member/ · Firebase Auth (member role)*

Mobile-first experience — the member's personal app for the club.

7.1 Dashboard

- Welcome greeting + membership status badge (Active / Low Credits / No Plan).
- Shake balance card with progress bar.
- Monthly wallet stats + recent transactions.
- Announcement ticker.
- Early Bird photo upload (when promotion is live).
- One-tap wallet top-up request.

7.2 Profile

- Edit info, upload avatar, change password.
- Personal stats (DOB, current / target weight, health conditions).
- Per-device push notification toggle.

7.3 Wallet

- Balance in the club's configured currency (e.g. Coins).
- Top-up request form with status banner.
- Transaction history + monthly spending chart.

7.4 Check-In

- QR scan (club QR or member-card QR).
- Manual ID fallback.
- Early-bird bonus when checked in before threshold time.

7.5 Attendance

- Monthly calendar with attended days.
- Streaks + lifetime visits + visual chart.

7.6 Today's Menu

- Daily shakes + Today's Specials.
- Photos, calories, price, in-stock badges.

7.7 Orders

- Place an order, watch its live status (Pending → Ready → Served).
- Order history with one-tap re-order.

7.8 Digital Member Card

- 3D flip animation: photo + ID front, balance + QR back.
- Download as PNG, native device share.
- QR scannable by reception & staff for check-in.

7.9 MLM Tree / Network

- Hierarchical visualisation of the entire downline.
- Direct & total network counts.
- Birthday / anniversary popup for network members.

7.10 Progress

- Weight log with chart.
- Goal vs current target.
- PDF export of the progress report.

7.11 Leaderboard

- Visits / weight loss / BTC / streak rankings, this month or all-time.

7.12 Announcements

- All club messages with read state and unread badge.

7.13 Login & Forgot Password

- Email + password (Firebase) + reset email flow.

8. Super Admin Portal

Route: /superadmin/ · Hosted on the platform domain*

The platform-wide command centre that operates every club from one place.

8.1 Dashboard

- Total clubs, total members, platform-wide orders.
- Per-club cards with member count, order count and Firebase cost estimate (INR).
- Payment status badges (Paid / Due Soon / Overdue).
- Unread enquiries count.

8.2 Clubs — List, Create, Detail

- Search / filter clubs by status and creation date.
- Create a new club (owner email, name, country, tier).
- Detail view: plans, members, billing, danger zone (archive / reset).
- Bulk import clubs from CSV.

8.3 Convert to Club

- One-click promotion of an enquiry into a fully provisioned club.
- Pre-populates from enquiry data and assigns subscription tier.

8.4 Platform Tree

- Global MLM structure across all clubs.
- Parent ↔ child relationships and roll-up performance.

8.5 Enquiries

- Lead inbox with status pipeline (New / Qualified / In Progress / Won / Lost).
- Comment history, email follow-up, convert-to-club action.

8.6 Global Menu

- Master menu templates pushed to all clubs.
- Clone-to-clubs bulk action; clubs may override locally.

8.7 Platform Settings

- Branding, email templates, subscription-tier configuration.
- FCM VAPID + Firebase config; feature gates per tier.

8.8 Upgrade Requests

- Free → Trial → Paid approval workflow with audit trail.

9. Cross-cutting Platform Features

Capabilities that run across every portal and tie the whole platform together.

Authentication & Authorization

- Firebase Auth with custom-claims role gating (member, staff, clubOwner, superAdmin).
- 6-digit PIN for kitchen / reception, 8-digit for owner.
- Domain-based tenancy (club domain vs super-admin domain) prevents cross-tenant leakage.

Push Notifications (FCM)

- Per-device token registration with automatic cleanup.
- Triggers: order ready, top-up approved / rejected, announcements, early-bird, fun-friday.
- Foreground (toast) and background (native) handling with deep-link payloads.

Face Recognition

- face-api.js (TinyFaceDetector + Landmark68 + Recognition).
- Enrollment + liveness check for inventory / stock approvals.
- PIN fallback if face capture fails.

Receipt Printing

- RawBT Android intent (rawbt:base64,...) for ESC/POS thermal printers.
- Order receipts, daily summary, top-up denial slips, Afresh tokens.
- Branded header, auto-print countdown, manual reprint.

Real-time Sync

- Live Firestore listeners on orders, members, inventory, announcements.
- Offline cache with automatic reconnect-and-sync.

PWA & Offline

- Service worker, offline banner, install-to-home-screen.
- Four dedicated Capacitor APKs: Reception, Kitchen, Admin, Landing.

MLM / Referral Engine

- Tree visualisation in graph and list views.
- Upline / downline lookup; birthday + anniversary popups for network members.
- Configurable commission structure.

Wallet System

- Per-member balance + transaction ledger.
- Top-up workflow: request → approve → credit.
- Auto-deduct on order served; bulk credit; referral bonus auto-credit.

Daily Schedule Automation

- Auto-applies the weekly schedule on first page load each day.
- Time-based push (e.g. 6am early-bird reminder).

Reporting / Analytics

- Daily report, attendance analytics, revenue by date / item / plan.
- Drastic weight-change flagging.
- CSV exports across all reports.

Billing / Cost Tracking

- Per-club Firebase usage estimation in INR.
- Monthly invoicing via Cloud Functions.
- Tier management (Free / Trial / Paid) with feature limits.

Mobile-first Responsive UI

- Mobile / tablet / desktop layouts.
- Bottom sheets on mobile, side panels on desktop.

10. Technical Capability Summary

Capability	Implementation	Status
Multi-tenant SaaS	Club detection by domain	Live
Firestore real-time DB	Live listeners + batch writes	Live
Firebase Auth	Email/password, custom claims, RBAC	Live
Cloud Functions	Notifications, cost estimation, jobs	Live
FCM Push Notifications	Tokens, broadcast, deep linking	Live
Face Recognition	face-api.js, liveness, PIN fallback	Live
Thermal Printer	RawBT intent, ESC/POS, base64	Live
MLM / Referral Tree	Parent-child, upline/downline viz	Live
Attendance Tracking	Check-in/out, hours, calendar	Live
Inventory Management	Unit-based, restock, recipe cost	Live
Wallet & Payments	Ledger, top-up, auto-deduct	Live
Digital Member Card	Flip card, QR, PNG export	Live
PWA & Offline	Service worker, offline indicator	Live
Dedicated Mobile Apps	4 APKs via Capacitor	Live
Reporting & Export	Daily reports, CSV export	Live
Early Bird / Fun Friday	Time-based bonuses, contests	Live
BTC Challenge	Multi-session, rankings, import	Live
Billing & Cost Tracking	Firebase usage, invoices, tiers	Live

Built for scale. Magic Nutrition Hub is a single platform that compresses what would normally take five separate systems — reception, kitchen ops, member app, owner CRM, and franchise HQ — into one real-time, mobile-first product.